Initial EDA Summary – Room Occupancy Detection

Dataset: Occupancy.csv · 20,560 rows · 2015-02-02 to 2015-02-18 · one reading per minute

20,560	0	23.1%	0.92
rows	missing / duplicate rows	of time occupied	Light–Occupancy correlation

1. Descriptive statistics

• The data is clean: no missing values and no duplicate rows, so no cleaning is needed.

• Light is heavily skewed: its median is 0 but its maximum is about 1,700, because the room is dark most of the time.

• HumidityRatio values are very small (~0.004) and need 4+ decimals to display correctly.

2. Occupancy distribution

• 76.9% Not Occupied vs 23.1% Occupied: the classes are imbalanced.

• A model that always predicts 'Not Occupied' already scores ~77% accuracy, so evaluate models with precision, recall and F1.

3. Sensors by occupancy

• **Light** jumps from 25 to 482 on average when people are present: the clearest signal.

• **CO2** rises from 605 to 975 ppm and **Temperature** by about 1.4 °C when occupied.

• **Humidity** barely changes (27.5 vs 28.1) and is weak on its own.

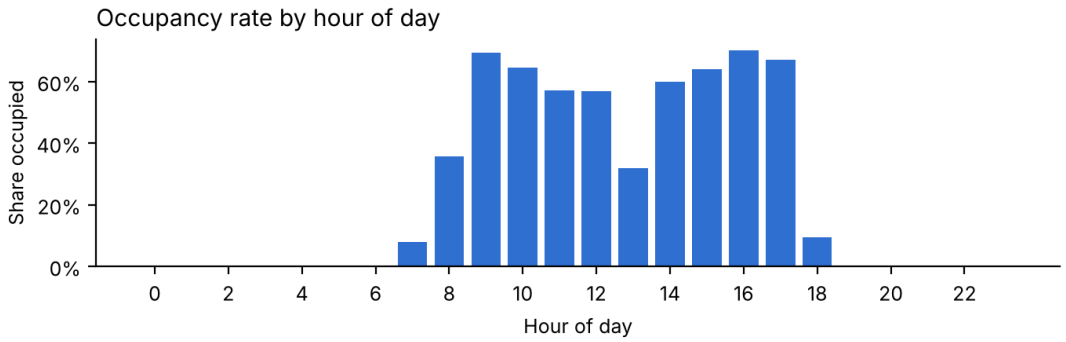
4. Patterns and relationships

• Correlation with Occupancy: Light 0.92, Temperature 0.56, CO2 0.50, HumidityRatio 0.26, Humidity 0.05.

• Humidity and HumidityRatio are almost duplicates (correlation 0.93).

• Occupancy follows working hours (~07:00–18:00) with a dip at 13:00 (likely lunch), and is zero on weekends.

• 832 rows have bright light (> 200) in an empty room, likely daylight or lights left on; these are the cases a light-only model will get wrong.



Conclusion

Occupancy is best detected by **Light**, supported by **Temperature** and **CO2**, and it follows a clear weekday working-hours routine. For modeling, keep these three sensors, treat Humidity/HumidityRatio as redundant, consider adding hour and weekday features, and use metrics that handle class imbalance.